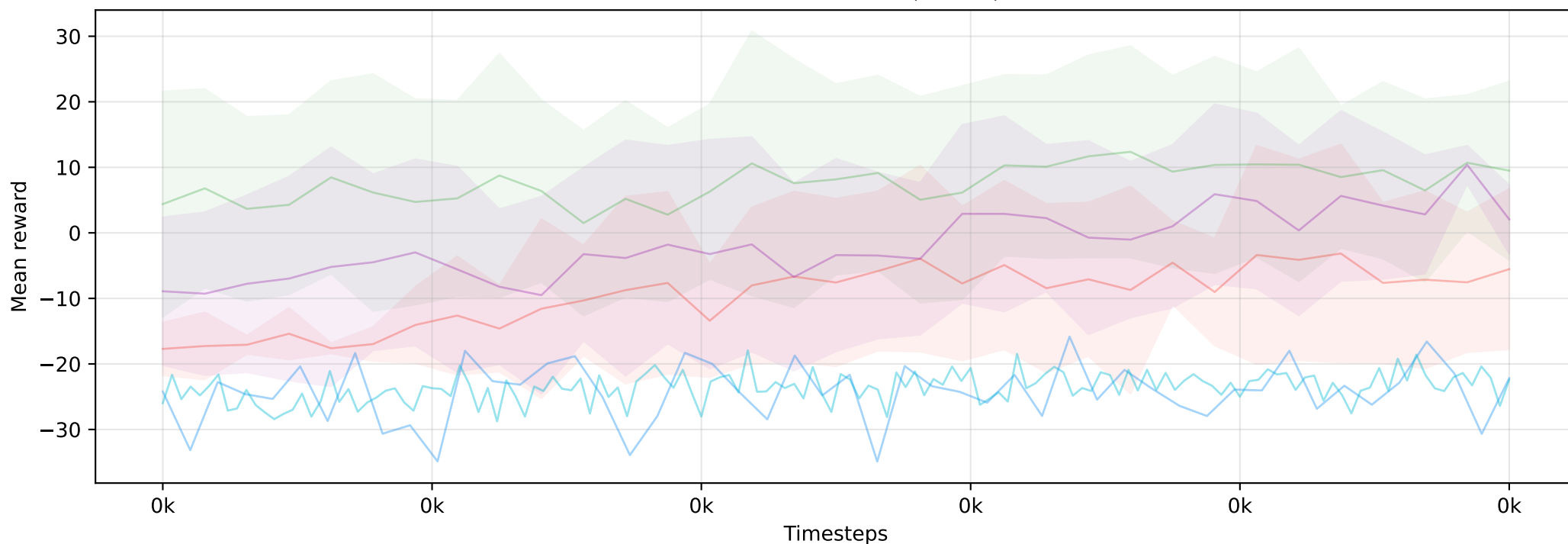
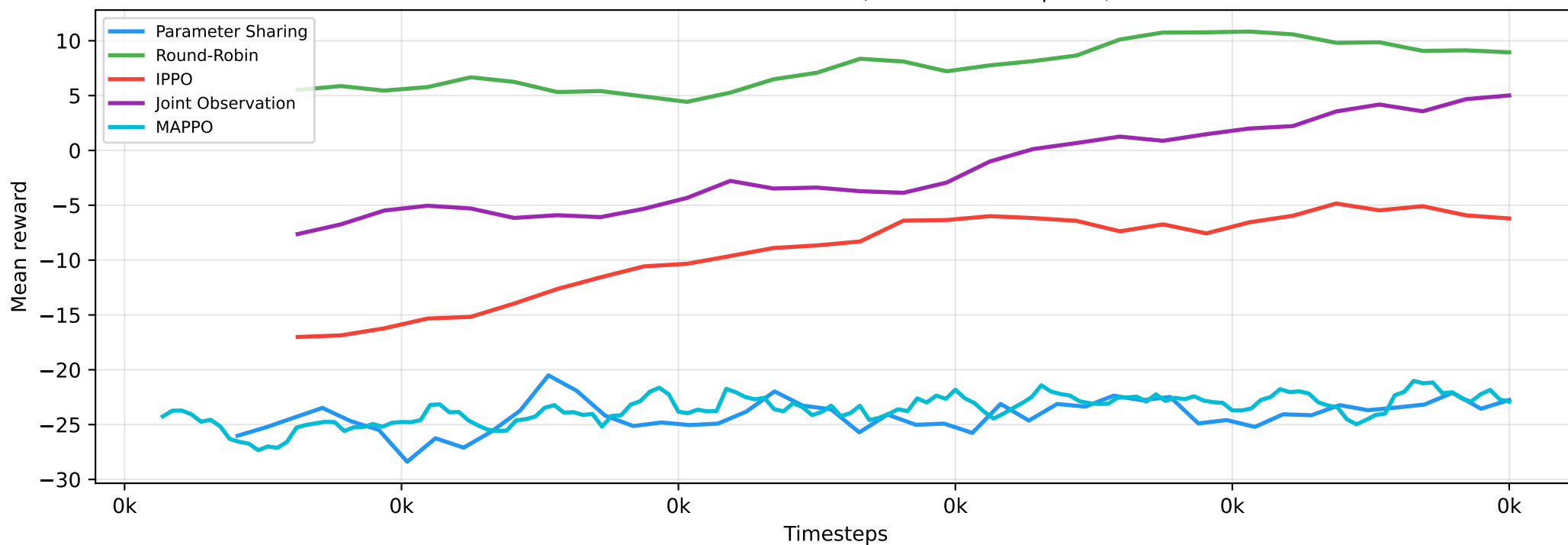


# PPO Evaluation — Mean Episode Reward

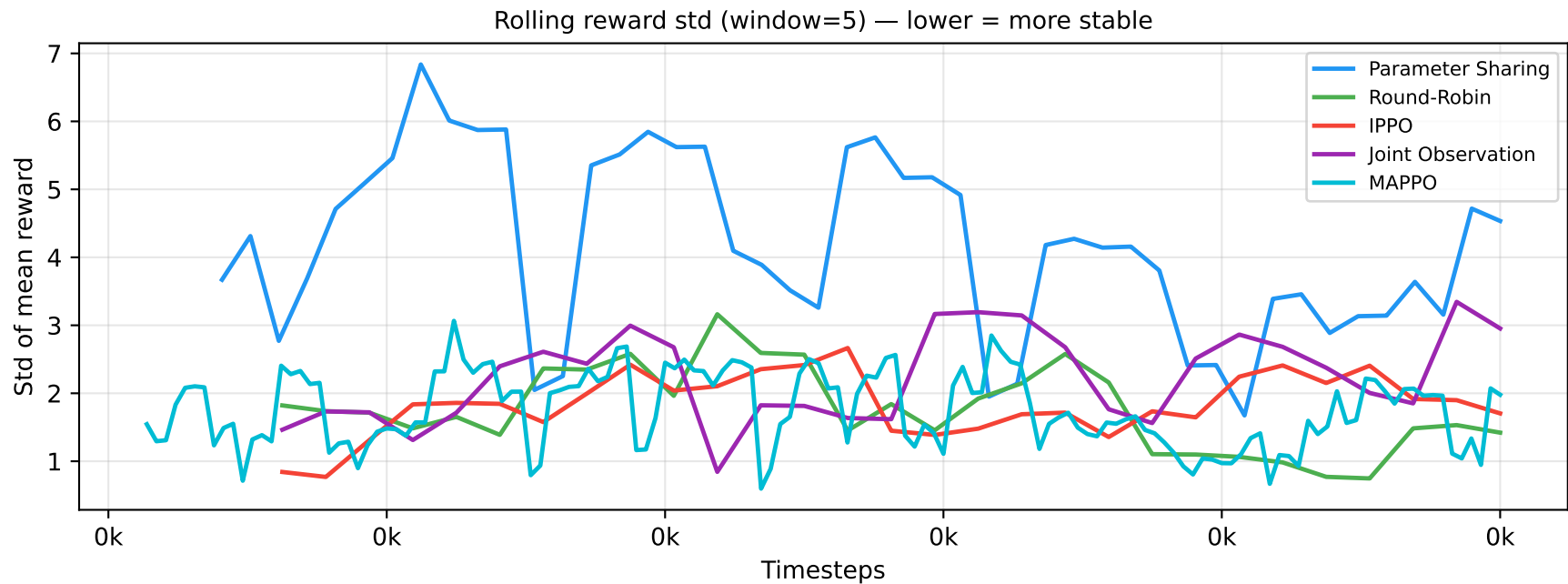
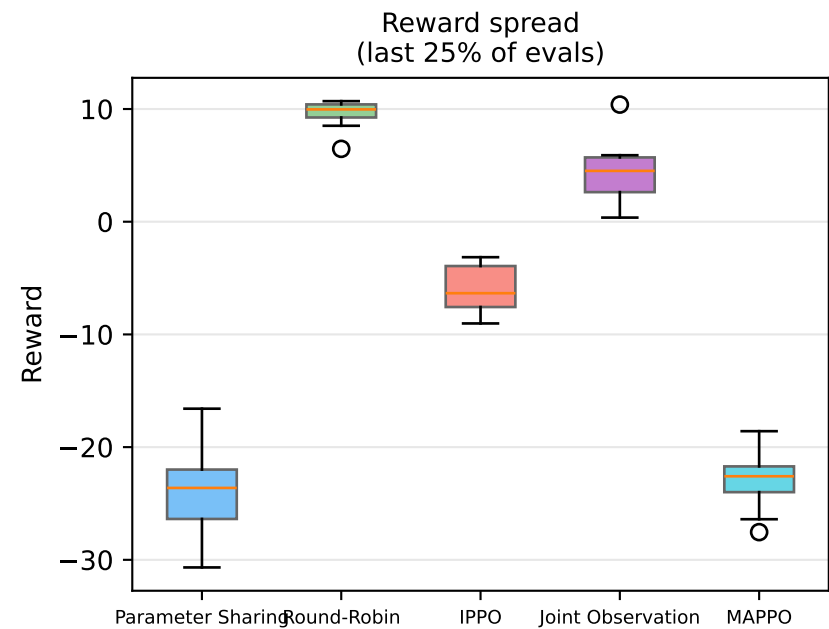
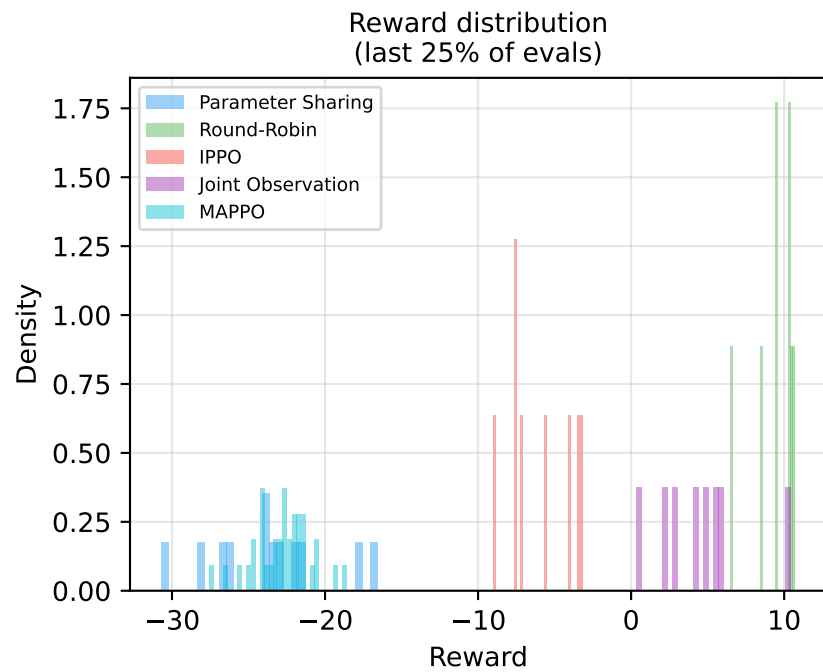
Raw eval mean reward ( $\pm 1$  std)



Smoothed eval mean reward (window=5 checkpoints)



# PPO Evaluation — Distribution & Convergence



## PPO Evaluation — Summary Statistics (last 25% of evals)

| Run               | Total steps | Eval checkpoints | Mean reward | Std   | Max     | Min     |
|-------------------|-------------|------------------|-------------|-------|---------|---------|
| Parameter Sharing | 0k          | 50               | -23.669     | 3.814 | -16.590 | -30.675 |
| Round-Robin       | 0k          | 33               | 9.494       | 1.327 | 10.698  | 6.465   |
| IPPO              | 0k          | 33               | -5.942      | 2.070 | -3.148  | -9.026  |
| Joint Observation | 0k          | 33               | 4.525       | 2.831 | 10.394  | 0.366   |
| MAPPO             | 0k          | 146              | -22.790     | 1.831 | -18.586 | -27.542 |